TEXAS A&M UNIVERSITY

MEEN 345 Fluids Lab *FALL 2025*

Title of Experiment: **Measurement and Analysis of Drag in Student-Created Car Models**	
Lab Section: 501	**Group #: 5**

All authors have contributed to the preparation of the report and have read the final version of the report.

On my honor, as an Aggie, I have neither given nor received unauthorized aid on this academic work.

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ABSTRACT

This study investigates the aerodynamic drag performance of a custom designed prototype of a vehicle model to explore principles of drag minimization relevant to automotive design. The primary objective was to design and fabricate a model with a fixed volume of 800 cm^3 that minimizes the drag force when subjected to fluid flow in a wind tunnel. The model was created using CAD software, 3D printed and tested across a range of Reynolds numbers by varying the upstream air velocity from approximately 3-35 m/s. Drag force was measured using force balance and was given by the program VDAS. The corresponding drag coefficients (C_d) and Reynolds numbers were calculated using the formula given in Lab 7. The analysis identifies key aerodynamic features of the model and evaluates their effectiveness. Conclusions discuss the relationship between observed drag, Reynolds number and model geometry, offering specific recommendations for shape optimization to reduce drag. This lab shows the significance of aerodynamic testing in scale model development, demonstrating a direct methodology for improving vehicle performance and efficiency through multiple prototypes and designs.

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NOMENCLATURE

Variable	Definition	Units
A	Area	m^2
C_D	Coefficient of Drag: quantifies the drag force experienced by an object in fluid flow	
F_x	Drag Force: component of force parallel to fluid flow	N
F_y	Lift Force: component of force perpendicular to fluid flow	N
g	gravity	$\frac{m}{s^2}$
h_{open}	The height of the manometer that is open to atmospheric pressure	mm
h_{tap}	The height of the manometer attached to the pressure tap	mm
P_{abs}	Absolute Pressure	kPa
P_{atm}	Atmospheric Pressure	kPa
R	Specific Gas Constant	$\frac{J}{kg*K}$
Re	Reynolds Number;	
U_∞	Free fluid stream velocity	$\frac{m}{s}$
μ	absolute/kinematic viscosity	$\frac{kg}{m*s}$
ρ_a	Density of air	$\frac{kg}{m^3}$
ρ_w	Density of water	$\frac{kg}{m^3}$

INTRODUCTION

Aerodynamic drag is a fundamental consideration in automotive engineering, directly influencing fuel efficiency, performance, and emissions in modern vehicles. To achieve optimal performance and design, engineers have to iteratively test prototypes, often starting with scale model analysis in controlled wind tunnel environments. This lab project explores this essential engineering practice by instructing us students to design, fabricate, and experimentally evaluate a car model with the primary objective of minimizing aerodynamic drag. The model was made under specific geometric and volume constraints, refined using CAD (SOLIDWORKS) software, and then fabricated using PLA 3D printing. Our group decided to change our design since we thought we could minimize the drag further, and then tested it in SOLIDWORKS, and then finalized printing this final model. This final model was then tested in a wind tunnel, where drag force measurements were calculated using a force balance and given to us by the VDAS program. Using this data and the ambient parameters, we then calculated the drag coefficients and Reynolds number, providing a quantitative assessment of the design's aerodynamic efficiency. This report details the model's construction, the experimental methodology, and the comprehensive analysis of our results and offers practical recommendations for design improvement and experimental refinement.

EXPERIMENTAL METHODS

The primary objectives of this lab revolved around deriving various flow values from drag data obtained from operating a wind tunnel over increasing velocities. The data was then used to gain an understanding of how Coefficient of Drag changes with increasing flow rates. In the wind tunnel, the group attached the additively manufactured car model to the test stand inside an acrylic observation chamber, supported by a metallic tube. This car is depicted in **Figure 1**, and a picture of the car in the context of the testing setup is seen in **Figure 2**.

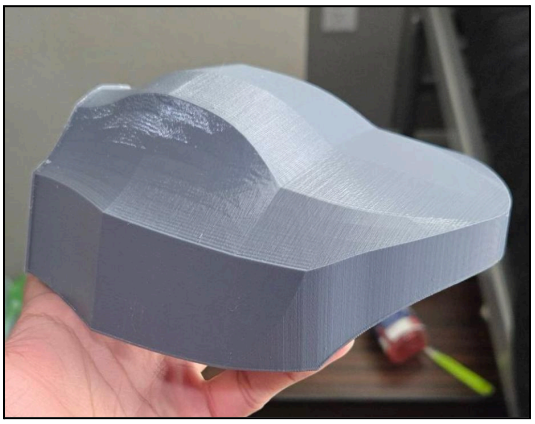


Figure 1. *The additively manufactured car model*

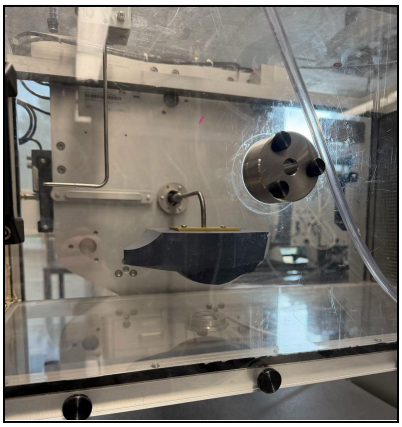


Figure 2. *A picture of the Wind Tunnel observation box with the car model mounted*

After screwing in the acrylic panel to the observation box, environmental barometric pressure and temperature was measured with a wall-mounted gauge seen in the testing room (**Figure 3**), before continuing with our lab. From there, the VDAS software was

booted up to mark the start of the data collection process. Starting at an intake velocity of 3 meters per second, the group incrementally increased the windspeed over 9 instances, measuring pressure tap head and average wall pressure head for each step using the pressure tap panel seen in **Figure 4**. Simultaneously, the VDAS system continued to measure the pressure drop across the two taps for each windspeed, as well as other valuable data such as contact forces. VDAS data will later be used to calculate important constants like the Drag and Lift Forces, as well as the Coefficient of Drag for our model.

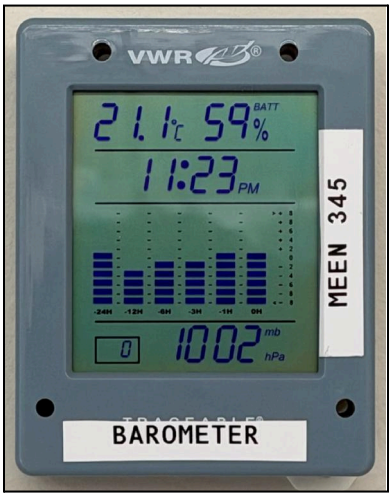


Figure 3. *Barometer with environmental readings on the day the experiment was conducted*

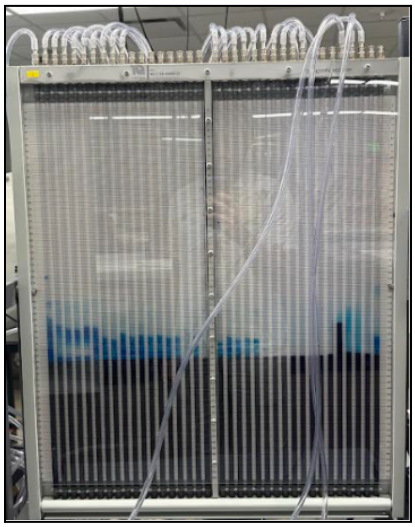


Figure 4. *The manometer that was used to determine the pressure head difference*

To better understand the flow around our car model, a smoke machine was also used to visualize the distinct streamlines in the acrylic observation box. However, when the machine was held up to the intake vent, the diffusion of smoke was too thin to see any unique flow behavior around the car model. Instead of this, our group used a simple computational fluid dynamics simulation in SOLIDWORKS to visualize the streamlines we would have seen in the acrylic observation box (**Figure 5**)

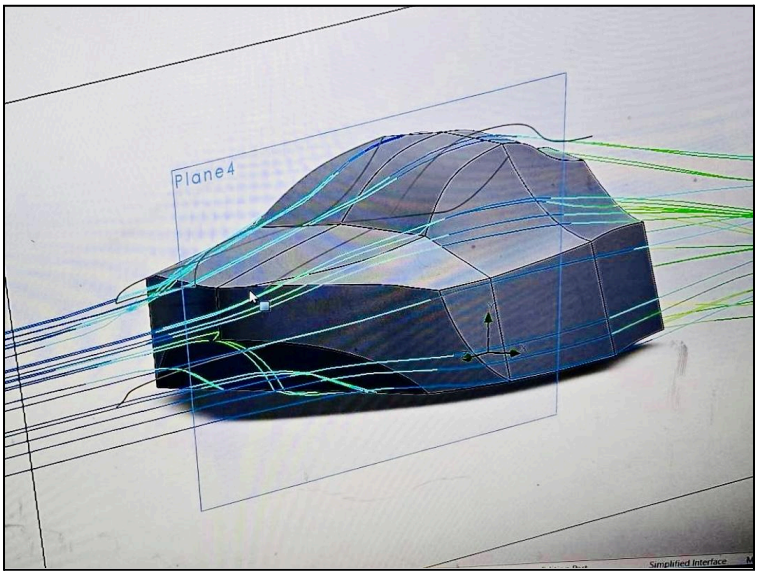


Figure 5*. CFD simulation of flow around car model*

RESULTS AND DISCUSSION

Measured and Given Constants:

Gravity (g) = 9.81 m/s^2

Density of water (ρ) = 997 kg/m^3

Room Temperature (T_{room}) = 300 K

Ambient Pressure P_{atm} = 100.3 ± 0.1 kPa

Specific Gas constant (R) $= 287.058 \frac{J}{(kg \bullet K)}$

Absolute/dynamic viscosity: $\mu = 1.8 \times 10^{-5} Pa \bullet s$

Contact Area *(Measured)* A = 0.00770999 m^2

Equations:

Convert Manometer height difference to pressure:

Gauge Pressure at Each tap = $P_{gauge,tap} = \rho_{water}(g)(h_{tap} - h_{open})$

Add gauge pressure to atmospheric pressure to find Absolute pressure.

Absolute pressure = $P_{abs} = P_{atm} + P_{gauge,\,tap}$

Use Ideal Gas Law with ambient conditions.

Air Density = $\rho_{air} = \frac{P_{abs}}{R*T_{room}}$

Calculate Drag (Fx) and lift (Fy).

Drag = $F_x = \sum F_x, k$, *Lift* = $F_y = \sum F_y, k$

Find coefficient of Drag.

Drag Coefficient = $C_d = \frac{F_x}{(.5*\rho*V^2*D*L)}$

Calculate flow regime using velocity, diameter and viscosity.

Reynold Number = *Re* = $\frac{\rho VD}{\mu}$

Normalize pressure difference by dynamic pressure

Coefficient of Pressure = $\frac{P_{abs,tap} - P_{abs,wall}}{.5 \times \rho_{air} \times V^2}$

Calculate drag force using ideal conditions.

Aerodynamic Force = $0.5 * \rho_{air} V^2 D L C_D$

Key Design Considerations:

Drag is a force heavily influenced by the magnitude of flow separation and static pressure difference across the surface of an object. As fluid moves around a body, it tries to stay attached to the surface and static pressure is converted to kinetic head through flow separation. If this separation gets large enough however, whether with high velocities or turbulent flows, the loss of static pressure gives rise to low pressure zones known as wake regions, which drastically increase drag forces. The larger this wake region, the larger the drag force acting on the object. For this reason, surfaces directly perpendicular to flow streamlines experience the maximum pressure differential due the formation of the largest possible wake region behind the contact area. With this in mind, the design considerations of our car model were centered around minimizing the contact angle between streamlines and the surface, allowing airflow to stick around our object and minimize flow separation as much as possible. This philosophy is reflected in the sleek, streamlined car design seen in **Figure 1 and 5**. One design constraint to point out, however, was the use of distinct radial segments instead of one smooth body. While this is less favorable to the aerodynamics of the model, it did make the part easier to manufacture with a 3D printer. This is what made this tradeoff necessary. Once a design was finalized however, the model was exported to a 3D printer which additively manufactured the car model with PLA.

Discussion of Data Analysis:

Tap pressure, wall pressure, air density, and flow velocity were all calculated to derive the distribution drag forces, lift forces, and Coefficients of Drag for each of the 9 iterations of windspeed flow velocities in **Figures 6 and 7**.

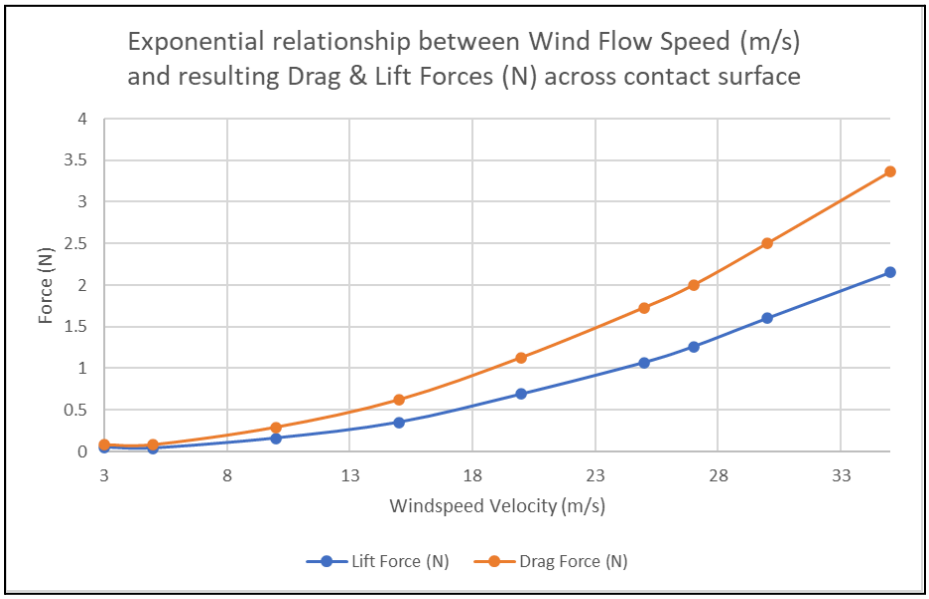


***Figure 6**. Correlation between Windspeed and the resultant Drag and Lift Forces for the*

In the results from putting the model car in the wind tunnel, **Figure 6** depicts a very distinct quadratic relationship between wind speed across the component and the corresponding lift and drag forces. This relationship likely derives from the aerodynamic force equation, where the resultant drag force is directly related to velocity squared. Because the angle of lift stays constant for the model car (setup does not change), lift force also follows a similar quadratic relationship with air flow velocity.

In **Figure 7**, as the Reynolds Number (Re) increases, the Coefficient of Drag appears to remain stagnant. Because the Reynolds Number (Re) is directly proportional to velocity, it is reasonable to assume that Coefficient of Drag would appear to stay constant even as the flow velocity increases as well. With moderate Reynolds Numbers like those measured from the windspeed range of 10-35 m/s in our experiment, it is likely that boundary layer thickness remains constant, flow separation is unchanging, and wake regions stay fixed. Because flow is not dominated by any overpowering, intermittent behaviors like turbulence or velocity, the Coefficient of Drag is able to remain constant for the flow speed range we selected.

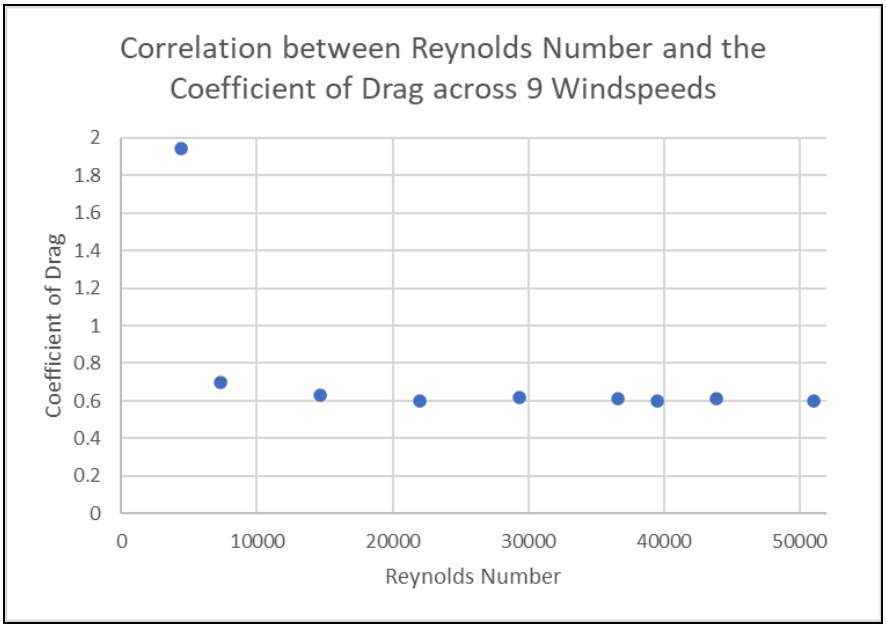


***Figure** 7. CFD simulation of flow around car model*

If we were to fall out of the moderate Reynolds Number range, as seen with low wind speeds like 3-5 m/s, viscous effects dominate. Because the flow is much more viscous at low speeds, it causes thicker boundary layers, resulting in early flow separation and larger wake regions. For these lower velocities, we would expect a higher Coefficient of Drag as seen in **Figure 7**. As velocity increases, Reynolds Number increases and turbulent behavior dominates, causing the sharp decline and then the stagnation of the Coefficient of Drag. It is likely that as turbulent behavior continues to dominate, the flow separation gets pushed further back on the surface of the model car, hypothetically causing a further decline in the Coefficient of Drag.

For many of these reasons, the most ideal vehicle would be an elongated teardrop shape if all manufacturing constraints were ignored. This is because it has shallowest contact angles with streamlines, allowing for flow separation to occur as far downstream as possible. Flow is able to attach to the surface and static pressure decreases and increases smoothly around the body. Because this design minimizes flow separation, there is no wake region to contribute to drag- effectively making this the most aerodynamic shape.

CONCLUSIONS

The specific problem being addressed in this investigation was how increasing flow velocity around a body impacts the Coefficient of Drag. To achieve an understanding of this behavior, the group designed our idea of an aerodynamic model car, minimizing large contact angles that would lead to flow separation, large wake regions, and higher overall drag. This model was additively manufactured and then placed in a wind tunnel, which subjected it to 9 instances of increasing flow velocities. A data acquisition system measured different features of flow, which allowed us to derive a correlation between Lift & Drag force and velocity in **Figure 6**. These results showed that aerodynamic forces had a direct quadratic relationship with velocity, increasing and decreasing with a square factor. More importantly, however, **Figure 7** showed that the Coefficient of Drag remains constant in moderate Reynolds Number flows, but is significantly higher in low Reynolds Number flows. This is because viscous-dominated fluids (with low Reynolds numbers) occur at lower velocities, and the thicker boundary layers are prone to early flow separation and larger wake regions. This is why we see the Drag Coefficient start out high and then decline sharply before stagnating for moderate Reynolds Number flows. Faster flows push back the flow separation and lead to small wake regions and therefore less drag. Future work could investigate how well these flow relationships carry over to a more ideal shape, such as the teardrop mentioned above, which would theoretically minimize flow separation so significantly that it is the most aerodynamic shape.

REFERENCES

[1] Kim, Haejune. "Pressure Distribution on a Cylindrical Cylinder." J. Mike Walker '66 Department of Mechanical Engineering

APPENDICES

A. Sample Calculations

Convert Manometer height difference to pressure:

Gauge Pressure at Each tap = $P_{gauge,tap} = \rho_{water}(g)(h_{tap} - h_{open})$= $997 * 9.81 * (.489 - .49)$=-9.8 Pa

Add gauge pressure to atmospheric pressure to find Absolute pressure.
Absolute pressure = $P_{abs} = P_{atm} + P_{gauge,tap}$ =100200+(-78.2)=100190 Pa
Use Ideal Gas Law with ambient conditions.
Air Density = $\rho_{air} = \frac{P_{abs}}{R*T_{room}}$ =$\frac{100200}{(287.058)/294.85}$=1.186
Calculate Drag (F_x) and lift (F_y).

Drag = $F_x = \sum F_x$, k= 0.08 N

Lift = $F_y = \sum F_y$, k= 0.05 N

Find coefficient of Drag.
Drag Coefficient = $C_d = \frac{F_x}{(.5*\rho*V^2*D*L)}$ =1.94
Calculate flow regime using velocity, diameter and viscosity.
Reynold Number = Re = $\frac{\rho VD}{\mu}$=$.4411808177 \times 10^4$
Normalize pressure difference by dynamic pressure
Coefficient of Pressure = $\frac{P_{abs,tap} - P_{abs,wall}}{.5\times\rho_{air}\times V^2}$=.08
Calculate drag force using ideal conditions.
Theoretical Drag Force = $.5 \times \rho_{air} * V^2 \times D \times L \times C_D$= 4.78 N

Table 1. *Uncertainty Values for Uncertainty Analysis*

Parameter	Value	Uncertainty
Drag Force	1.13 N	N/A
Fluid Density	1.183 kg/m^3	0.001 kg/m^3
Wind Speed	20 m/s	0.1 m/s
Area of Contact	$7.8*10^{-3}$	N/A

The following uncertainty calculations were done using our data collected from a wind speed of 20 m/s. The uncertainty analysis we will be performing will be on the coefficient of drag, which is a value that quantifies the drag force experienced by our car against the fluid flow of the air moving against it. We use the equation below to get our calculated number.

$$C_D = \frac{F_x}{0.5\rho U_\infty^2 A}$$

Where ρ is our fluid density, U_∞ is our wind speed value, F_x is our drag force, and A is our area of contact. When inputting our numerical values, we find a C_D of 0.62 for our model car in a 20 m/s environment.

Our uncertainty equation gives us the following:

$$\sqrt{\left(-\left(\frac{0.5F_x}{(0.5\rho U_\infty^2 A)^2} * A * U_\infty^2\right) * \delta\rho\right)^2 + \left(-\left(\frac{2F_x}{(0.5\rho U_\infty^2 A)^2} * 0.5 * A * \rho * U_\infty\right) * \delta U_\infty\right)^2}$$

This equation gives us an uncertainty value of 0.00615 when inputting the numbers from our 20 m/s wind speed trial.

B. Raw Data

Table 2. *Wind Tunnel Raw Data*

Velocity (m/s)	Drag Force (N)	Coefficient of Drag	Reynolds Number	Coefficient of Pressure	Theoretical Drag Force (N)
3	0.08	1.94	4411.808177	0	0.08
5	0.08	0.70	7352.295827	0	0.08
10	0.29	0.63	14698.84924	0.33	0.29
15	0.62	0.60	22028.89323	0.22	0.62
20	1.13	0.62	29328.78957	0.08	1.13
25	1.73	0.61	36607.15187	0.08	1.73
27	2.00	0.60	39508.59113	0.07	2.00
30	2.50	0.61	43842.44609	0.09	2.50
35	3.36	0.60	51033.95444	0.11	3.36

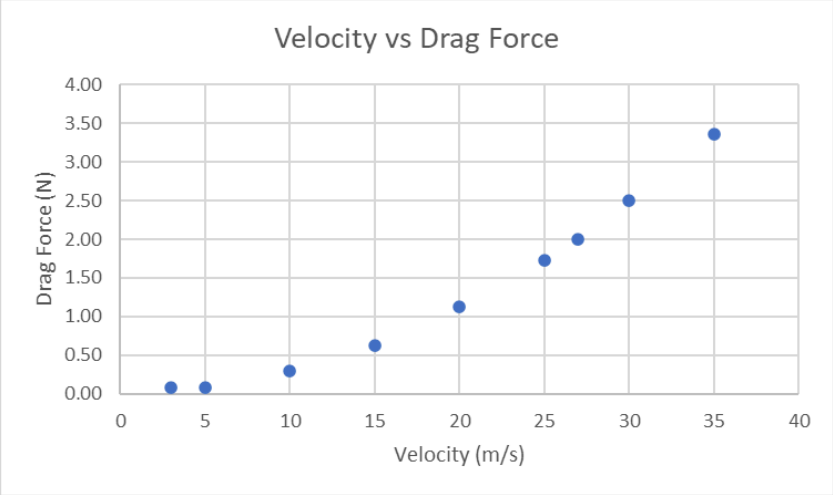


Figure 8*. Correlation between Velocity and Drag Force*

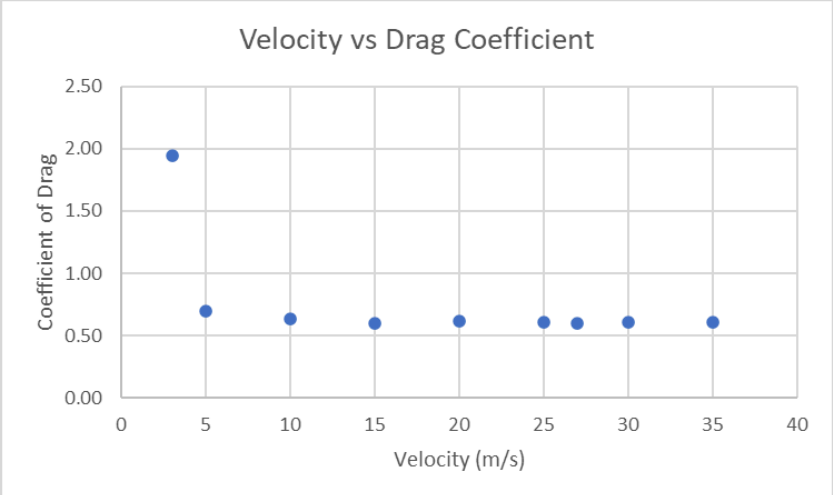


Figure 9*. Correlation between Velocity and Drag Coefficient*

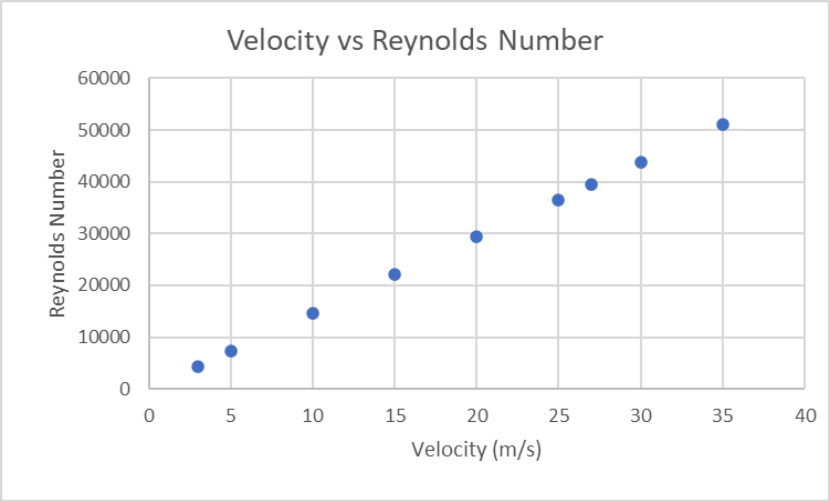


***Figure 10**. Correlation between Velocity and our Reynolds Number*

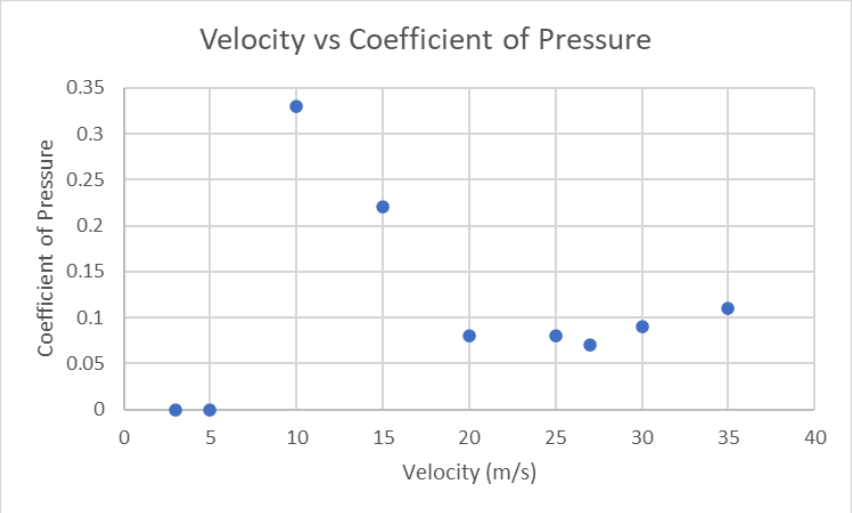


***Figure 11**. Correlation between Velocity and our Coefficient of Pressure*